# Campi Flegrei unrest: independent observational results and reproduction record

Independent reproduction of Tan et al. (2025)

12 October 2026

**Main paper**

**Abstract**

Original displacement observations and an explicitly permitted routine earthquake catalog reconstruct the historical context of unrest at Campi Flegrei. Daily GNSS observations now extend through 20 March 2025. Independent waveform processing produces 740 provisional GaMMA associations from 63036 picks, but has not established the publication's relocated fault geometry, magnitude distribution or focal mechanisms. No publication panel is fully independently reproduced. Partial counterparts: 1A, 1B, 1C, S1, S2, S5/map, S5/northing-depth, S5/easting-depth, S6/map, S6/northing-depth, S6/easting-depth, S8A, S8B, S15C. All remaining panels are not reproduced. The scientific narrative and reproduced supplement precede the input, implementation, validation and completion records.

Campi Flegrei's unrest is recorded by ground displacement and seismicity. Tan et al. (2025) use a machine-learning-assisted earthquake catalog to resolve active structures and evaluate possible sources of unrest. Their study is observational: its central results describe the location, timing and character of earthquakes, rather than solutions of a coupled physical forward model. Here, the completed independent calculations are placed in the same scientific sequence. Published images are confined to the separate comparison PDF.

## The current phase of unrest

Figure 1 reconstructs the historical context from the explicitly permitted routine catalog and original displacement observations. The routine snapshot contains 16,046 located events from 2005 through 20 March 2025, including five events with duration magnitude at least 4.0. Monthly counts include all routine events; the cumulative series includes 7,218 events with $M_d > 0.2$. These external records are separate from the independently detected catalog.

The displacement reconstruction contains 128 leveling entries and 8,743 daily GNSS observations through 20 March 2025. The original 1905–2023 interval is extended with observed data, and the extension is shaded. A common vertical datum is obtained from overlapping observation records. The long sequence resolves subsidence, the major twentieth-century uplift episodes, and the recent uplift. It does not by itself distinguish magmatic from hydrothermal forcing. All three panels remain partial because the historical catalog snapshot, basemap and exact displacement-stitching convention differ or are incompletely specified.

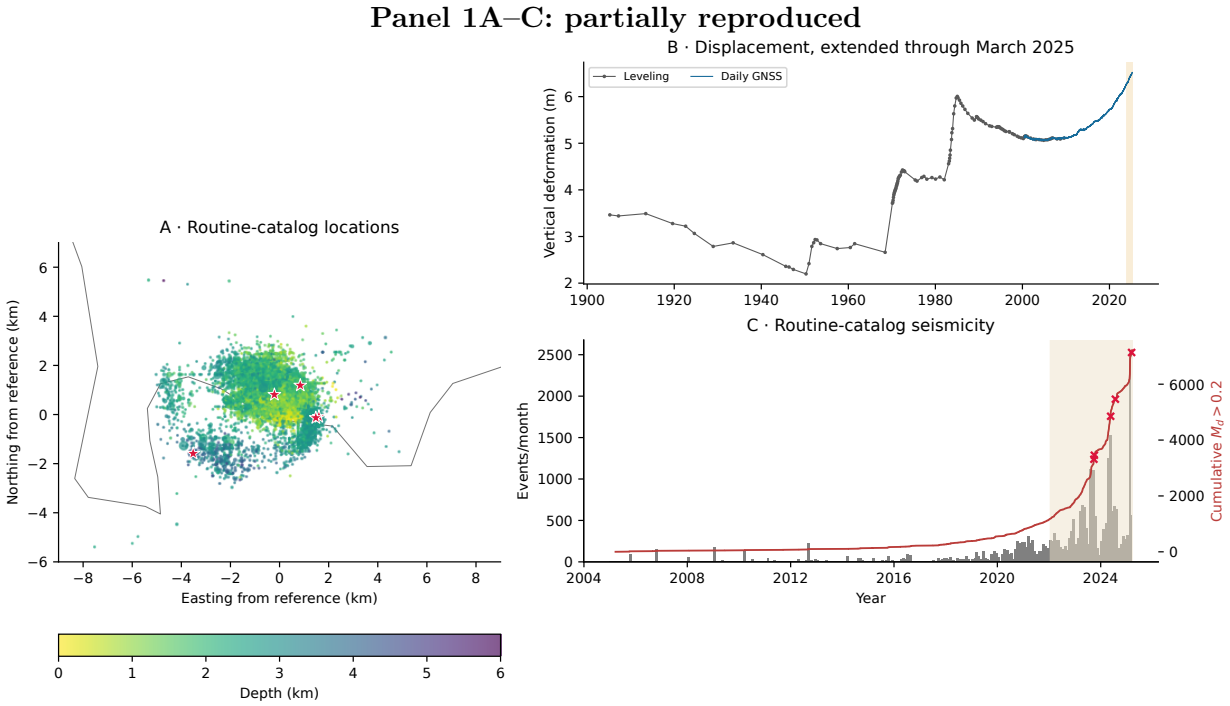



**Figure 1:** Seismicity and ground displacement. Partial reconstruction from permitted original displacement observations and external routine-catalog records. A: 16,046 located routine events and five Md>=4 stars. B: historical leveling datum; daily GNSS through 20 March 2025, with the extension beyond 2023 shaded. C: all monthly counts and exact-time cumulative Md>0.2; March includes days 1–20 only. This is not the independently detected catalog. The basemap and live catalog snapshot differ from the publication.

## Improving the earthquake catalog

Independent PhaseNet inference and GaMMA association yield 740 provisional events in the processed waveform sample. These are initial associations with an alternative starting velocity profile. A limited VELEST pilot is reported in the supplement, but regional calibration and double-difference relocation have not established a catalog comparable to Tan et al.'s result. Consequently, they cannot support an independent multiyear seismicity rate or depth–time distribution (Figure 2).

**All panels/views — not reproduced.** Panels A and B need multiyear calibrated/relocated events and magnitudes. The current 740 initial associations remain provisional.

**Figure 2:** Temporal evolution of seismicity. Seismicity rates by region and depth versus time are not reproduced.

## A clearer view of the current phase of unrest

Figures 3 and 4 address the publication's main structural observations: the distribution of seismicity and the source mechanisms of larger events. The present independent work has no calibrated, relocated catalog or focal-mechanism solutions. It therefore cannot determine the depth range, continuity or temporal activation of the ring fault, resolve the central fault system, or establish its slip orientations. The placeholders retain these scientific questions without inserting provisional diagnostic locations or supplied author solutions.

**All panels/views — not reproduced.** The map and both cross sections require independently calibrated and double-difference-relocated events. Provisional diagnostic locations are not substitutes.

**Figure 3:** Spatial distribution of seismicity. Map, northing-depth and easting-depth views are not reproduced.

**All panels/views — not reproduced.** Panels A and B require independent SKHASH/gCAP solutions and reliable locations. No focal inversion was executed.

**Figure 4:** Focal mechanisms of larger events. Ring-sector and central/onshore focal-mechanism maps are not reproduced.

## Implications for the physical process driving the unrest

The reconstructed hybrid-template example in Figure S15C is an independently processed waveform observation. The limited template search produced 702 unconfirmed threshold crossings; it does not establish recurring hybrid families or their evolution. Neither this result nor the initial associations can test the publication's inference about the presence or absence of upward seismic migration. The partial displacement history also does not distinguish magmatic and hydrothermal forcing. Such an interpretation would require the unresolved spatial and source constraints above.

## New perspectives for improved seismic risk assessment

Fault geometry and a calibrated magnitude distribution are needed to connect this analysis to seismic risk. The pilot magnitude measurements have not established the multiyear magnitude-frequency distribution required for Figure S8. The current results therefore provide no revised estimate of event probabilities, damaging-event locations or source dimensions. Their supported contribution is restricted to the partial observational reconstructions shown here and in the supplement.

## Reproduced supplement

This supplement follows the original Materials and Methods topic sequence, then Figures S1–S15 and Tables S1–S2. It reports only outcomes supported by the saved independent calculations. Detailed processing settings, input inventories, assumptions and validation are collected after this supplement.

## Materials and Methods

## S1 Building the Campi Flegrei high-resolution earthquake catalog

**Data.** The independently retrieved station intervals and station inventory provide partial counterparts to Figures S1 and S2. They do not establish complete network coverage over the study period.

**Phase picking.** Inference with released trained weights produced 63036 merged picks (8894 P and 54142 S) in the processed observations. This is inference, not reproduction of the training demonstrations S3–S4.

**Phase association.** GaMMA associated 5240 picks into 740 provisional events using the explicitly labeled published alternative starting profile. These are candidates, not a validated multiyear catalog.

**Initial earthquake location.** The calibration screen identified 98 candidates. The independent VELEST pilot completed 9 runs and retained 122 events after the final filters. The small calibration subset and absent independent phase review do not establish a reliable regional velocity model. The fitted layer $V_p/V_s$ ratios range from 1.214 to 2.076. The two regional starting choices yield RMS differences of 0.248 km/s in $V_p$ and 0.304 km/s in $V_s$. These sensitivities limit geological interpretation. Repeating calibration with 64-bit rather than 32-bit arithmetic changes some fitted layers by up to 0.48 km/s in $V_p$ and 0.60 km/s in $V_s$. This substantial arithmetic sensitivity prevents accepting the pilot fit as a robust regional calibration. Figure S6 shows a partial pilot counterpart when final quality-filtered events exist; it does not establish a validated regional model for Table S1. Figure S5 now presents the permitted routine-catalog reconstruction, with 12,503 located study-period events.

**Earthquake relocation.** Neither TomoDD relocation nor regional HypoDD SVD uncertainty estimation has been completed on an accepted independent catalog. Figures S7 and S9–S13 remain unreproduced.

**Local magnitude.** The pilot measured 12473 channel amplitudes; 2043 passed the 5 dB SNR threshold. It formed 377 relative-amplitude equations and returned 22 provisional magnitude estimates. Figure S8 displays the partial pilot comparison and distribution. This does not establish the final relocated catalog's magnitude distribution or completeness.

**Focal mechanism estimation.** SKHASH and gCAP inversions were not run. No independent mechanism is available for Figure 4.

## S2 Validation of earthquake locations

**Ring-shaped structures with confined depth distributions.** The provisional associations do not establish a ring structure. Internal travel-time consistency is preserved in the documentation, but does not substitute for the regional uncertainty or common-receiver tests in Figures S7 and S12.

**Depth of earthquakes in the central region.** There are now 40 associated CPOZ P arrivals in the provisional sample. The specified CPOZ waveform gathers and manual checks on apparently deep central events remain unperformed. No conclusion about a lower depth limit follows from their absence.

## S3 Template Matching to Detect Hybrid Events

The search covered 59.757 hours of available common CAAM/COLB intervals and returned 702 unconfirmed threshold crossings. It did not produce independently confirmed families for Figure S14.

**Depth.** No independent hybrid-event location set is available for Table S2.

**Frequency index.** Frequency indices were calculated for the retained three templates and the earlier four candidate windows. The expanded scan candidates have not received that spectral review. Figure S15C displays the corresponding third written-method template. The first two templates are different examples from the published S15A/B panels and are not inserted in their place.

**Relative magnitude.** Cluster-relative magnitudes were not assigned because cluster membership has not been established.

**Temporal evolution of the hybrid events.** The limited exposure and unconfirmed detections do not support a three-year cluster history.

**Further search for HB/LP events.** Catalog-wide frequency-index and coda-contamination review remains incomplete.

# Supplementary figures

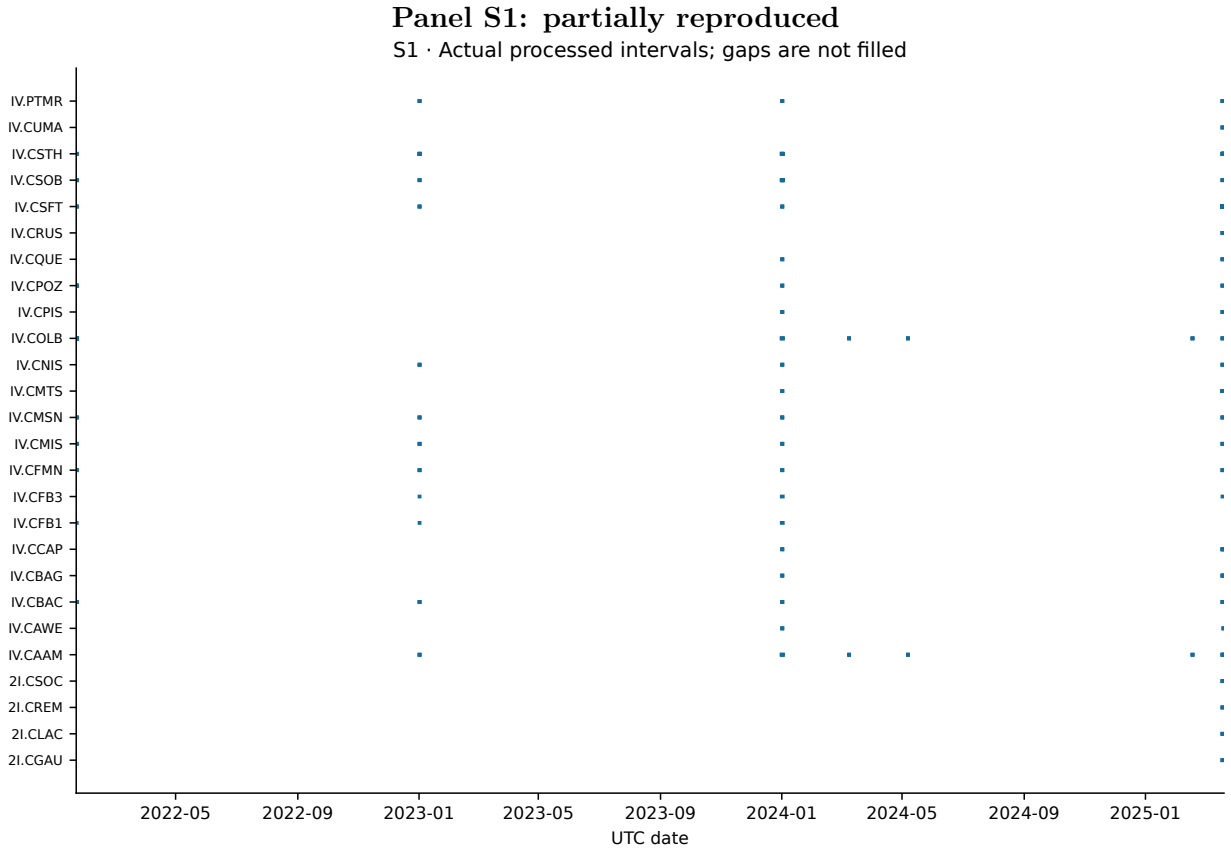



**Figure S1:** Recorded station availability. Partial: 318 processed files, 26 station names and 2955.824 channel-hours after union of overlapping intervals. Empty intervals are not connected or counted as coverage.

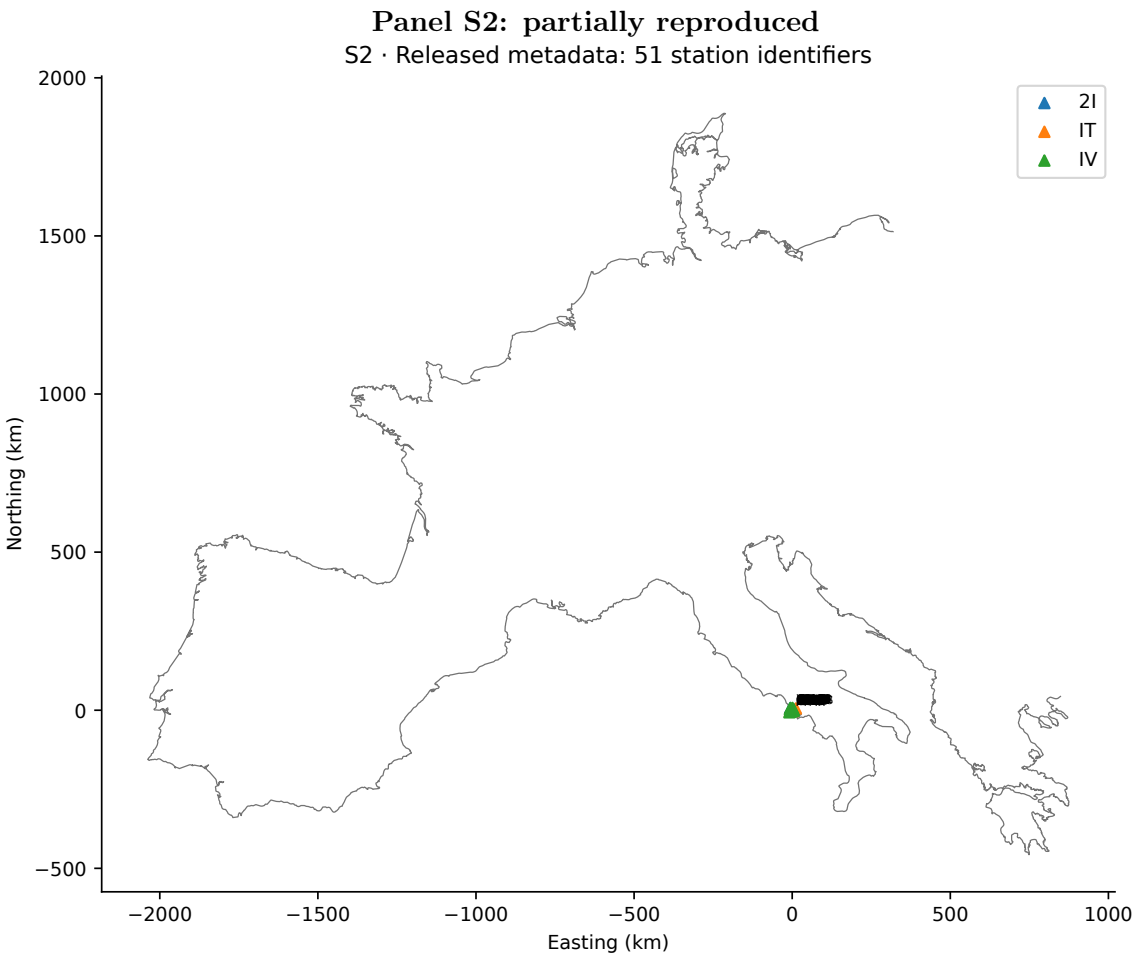



**Figure S2:** Station locations. Partial: released station coordinates and retrieved IV/2I response metadata. Exact membership of the original 48 instruments, station epochs and the basemap are not fully reconciled.

**All panels/views — not reproduced.** Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

**Figure S3:** Training examples. Noise, single-event and overlapping-event training examples are not reproduced.

**All panels/views — not reproduced.** Panels A and B were not generated for the exact CSOB swarm and CMIS noise examples. Other unlabeled inference comparisons remain diagnostics.

**Figure S4:** PhaseNet benchmark examples. The original benchmark examples and a labeled accuracy comparison are not reproduced.

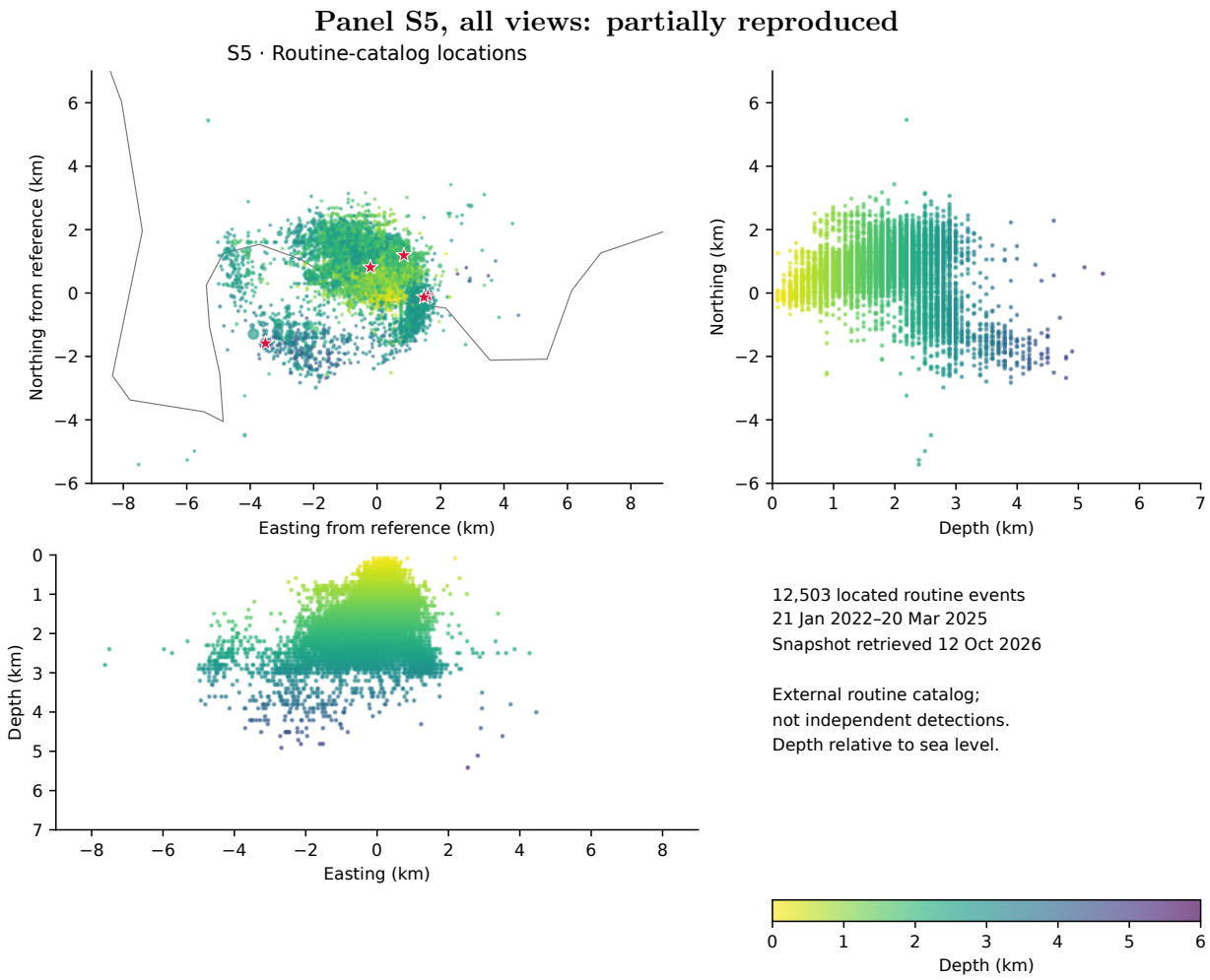



**Figure S5:** Routine-catalog locations. External routine-catalog data, independently selected and plotted: 12,503 located events, 21 January 2022–20 March 2025. This is 420 more than the published historical snapshot. No author-derived catalog is used to adjust that difference; these are not independent detections.

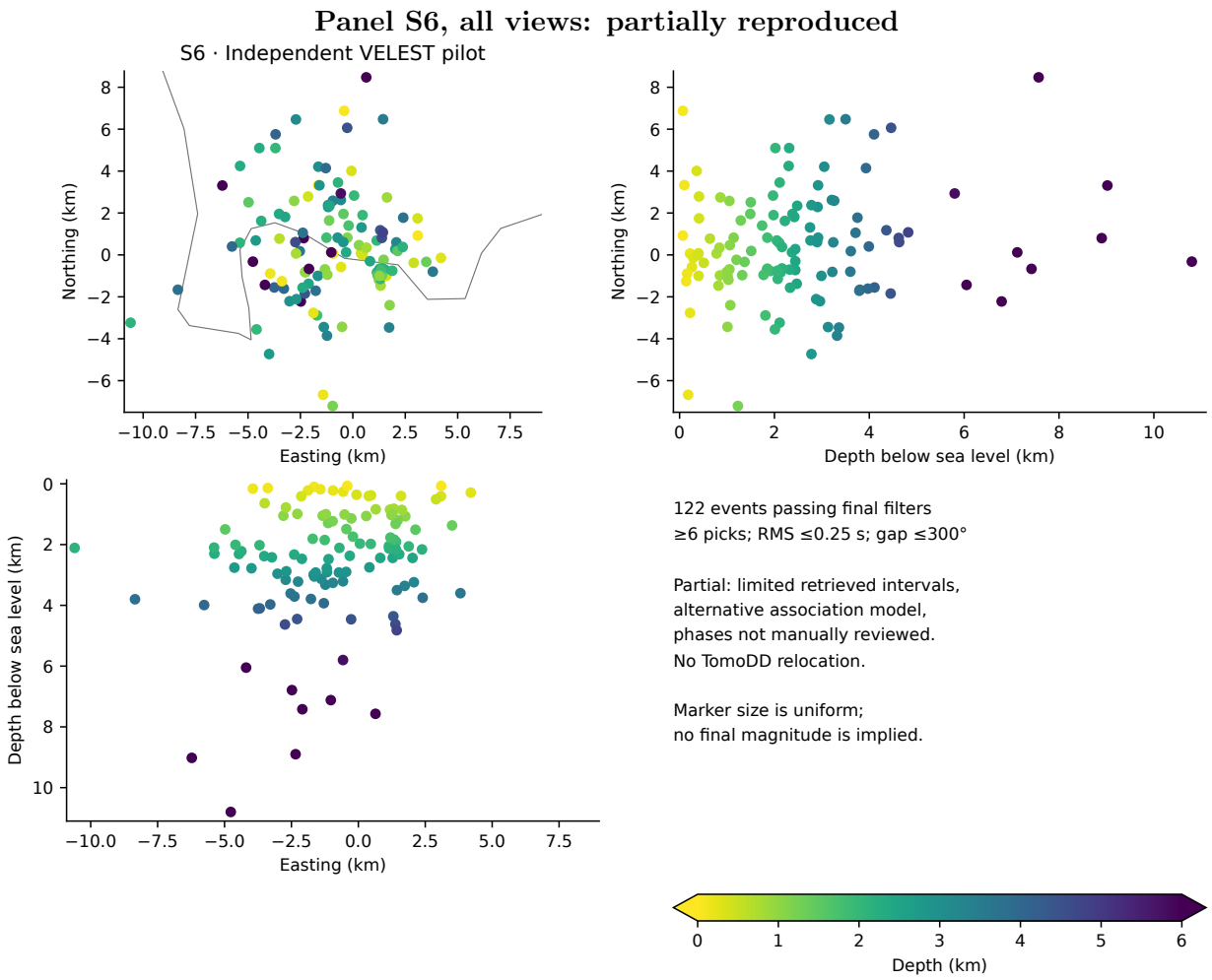



**Figure S6:** Initial VELEST locations. Partial independent VELEST pilot: 122 events pass the final six-pick, 0.25 s RMS and 300-degree gap filters. This limited, unreviewed sample is not the published 61,809-event catalog. The initial association model and regional averaging choices differ or are assumed; no TomoDD refinement is implied.

**All panels/views — not reproduced.** Panels A–G require independently selected regional subsets, HypoDD SVD locations and coordinate uncertainties. None of these scientific inversions ran.

**Figure S7:** Location uncertainties. Regional locations, uncertainty-versus-count panels and uncertainty distributions are not reproduced.

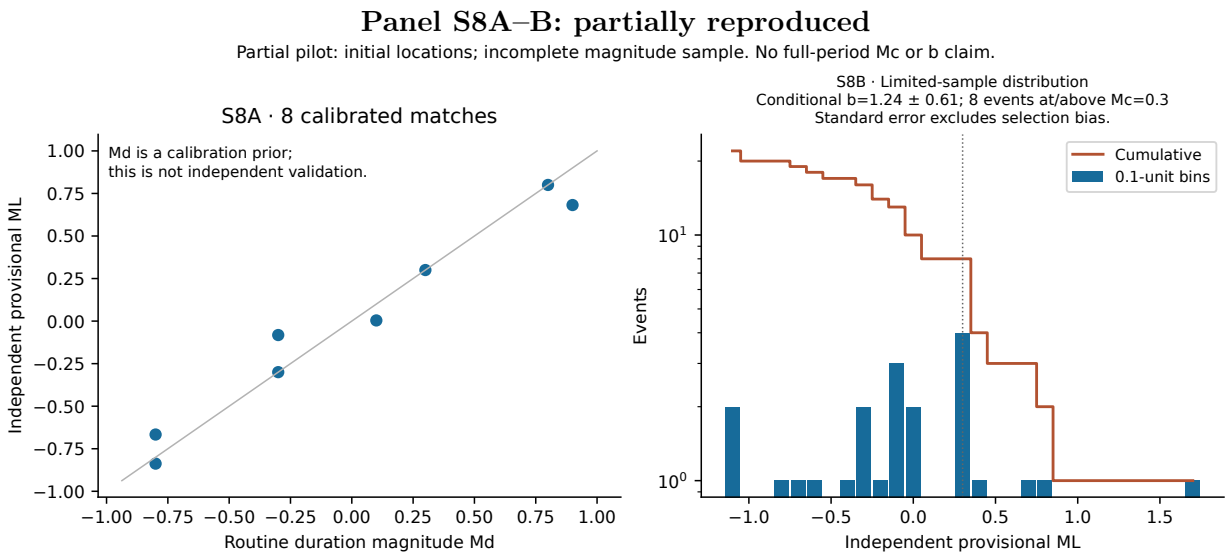



**Figure S8:** Magnitude calibration and frequency distribution. Partial amplitude-ratio pilot: 22 provisional magnitudes from independent observations, with routine Md as the documented calibration prior. A is not an independent validation of that prior. B is a limited-sample histogram with a conditional likelihood estimate, not the full-period completeness or b-value result. Locations are initial associations, before final relocation.

**All panels/views — not reproduced.** All three views require independently relocated events and cross-correlation observation counts for the published quality filter. These outputs are absent.

**Figure S9:** Best-located earthquake subset. The filtered map and cross sections are not reproduced.

**All panels/views — not reproduced.** The four annual groups for 2022–2025 require the independent multiyear relocated catalog. The available sample does not support them.

**Figure S10:** Seismicity by year. All four annual groups are not reproduced.

**All panels/views — not reproduced.** The four swarm maps require the independent relocated catalog and justified swarm windows. These selections and locations remain incomplete.

**Figure S11:** Four major swarms. All four swarm maps are not reproduced.

**All panels/views — not reproduced.** Panels A–H need relocated sector selections and manual depth review. 40 independent CPOZ P arrivals now exist, but the required sector gathers remain unfinished.

**Figure S12:** Ring sectors and CPOZ common-receiver gathers. Sector views and six arrival-aligned waveform gathers are not reproduced.

**All panels/views — not reproduced.** Panels A and B require reliable independent relocated coordinates and an appropriately sourced basemap. The relocated catalog does not exist.

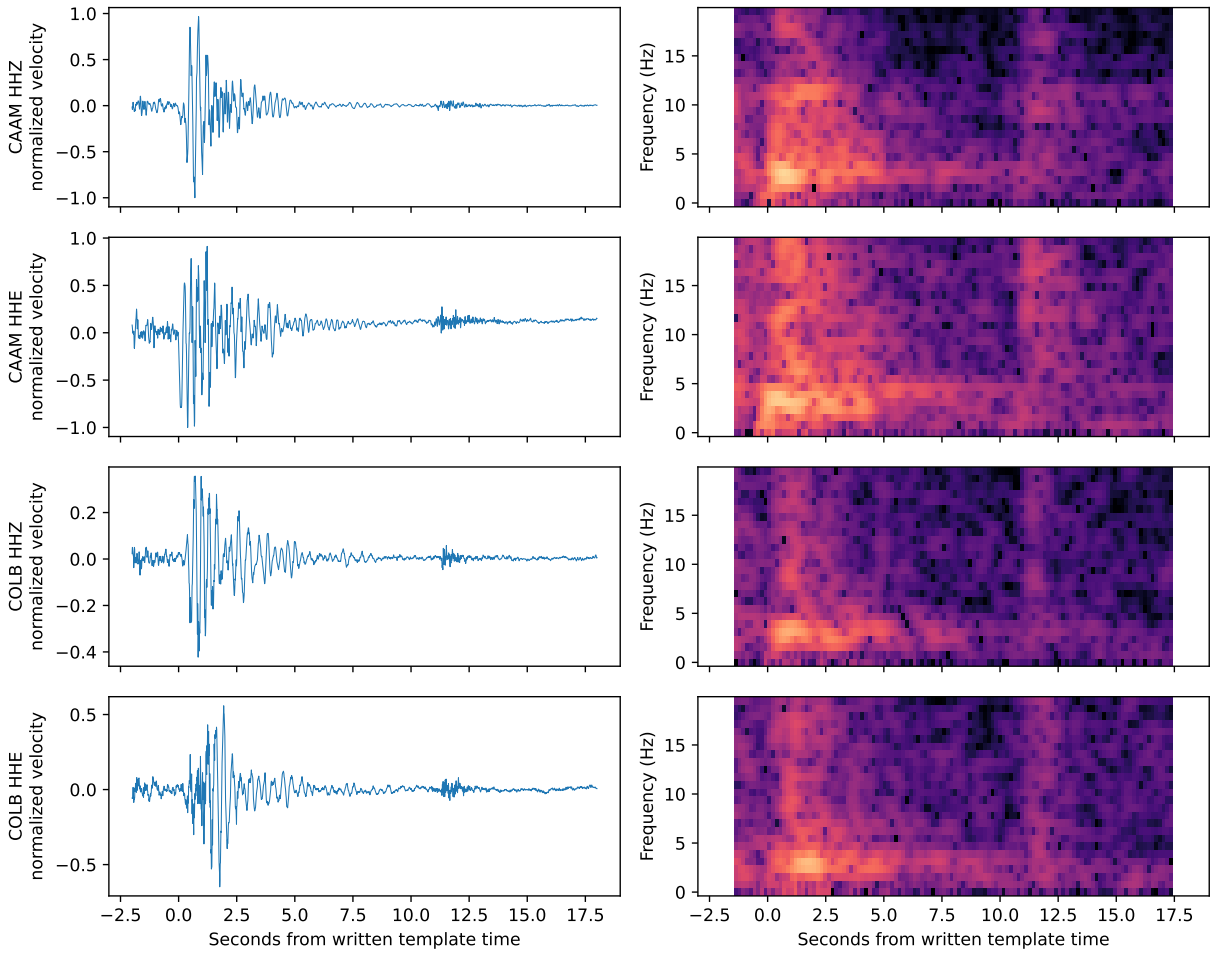
**Figure S13:** Relocated seismicity on satellite imagery. Full-caldera and onshore spatial views are not reproduced.

**All panels/views — not reproduced.** The cluster histories require three-year exposure, confirmed events and relative magnitudes. The present 59.757 hour scan produced 702 unconfirmed threshold crossings, not reviewed families.

**Figure S14:** Hybrid-event clusters. Cluster histories and annotated waveform collections are not reproduced.

**Panels A and B — not reproduced.** The retained E1/E2 templates are different published examples and are not substituted.

**Panel S15C: partially reproduced**




**Panel D — not reproduced.** No independently selected matching shallow-event example.

**Figure S15:** Hybrid and shallow-event waveforms. Panel C is partial: independently processed E3 CAAM/COLB observations. Onset alignment, display filtering and spectral settings differ or are incompletely specified. A, B and D have placeholders rather than substitute waveforms.

## Supplementary tables

**Table S1: fitted one-dimensional velocity model — not reproduced.** No independently validated regional VELEST model is accepted. Pilot fits and failed attempts, when present, are recorded after the supplement; the author's fitted velocities are not substituted.

**Table S2: hybrid events in the relocated catalog — not reproduced.** No independently relocated hybrid-event sample exists. The author's event parameters are not substituted.

## Supplementary data products

Independent counterparts of the final catalog (Data S1), associated phases of that final catalog (Data S2), focal mechanisms (Data S3), and the relocated interactive view (Interactive Plot S1) are not complete. The saved provisional products retain their existing labels. References and the complete provenance record follow in the reproduction documentation.

## Reproduction documentation

This record describes the current continuation, not the superseded nine-event run. Historical outputs remain recoverable from Git. The main paper and reproduced supplement contain only corresponding publication panels; published images occur only in the separate comparison PDF.

## D1 Scientific specification and input verification

The local main paper and supplement were read before changing scientific outputs. Main captions 1–4, supplementary captions S1–S15 and the methods are readable. The two uploaded ZIP archives were listed and CRC-tested without executing their contents. Their exact filenames, sizes and checksums remain in the input inventory.

| File | Bytes | Inspection |
|---|---|---|
| `tanmodel.pdf` | 2140554 | Text and captions readable |
| `tanmodelsupp.pdf` | 62164947 | Text and captions readable |
| `science.adw9038_data_s1_to_s3.zip` | 12960255 | Directory readable; CRC passed; not executed |
| `science.adw9038_interactive_plot_s1.zip` | 8852485 | Directory readable; CRC passed; not executed |

Data S1 is the authors' derived catalog, S2 their associated phases, and S3 their focal mechanisms. The second archive contains an interactive derived-result visualization. None supplies original waveform observations for the independent workflow. These products are excluded from initialization, filtering, calibration, tuning and plot filling. The previous comparison-only audit counted 74 mechanism rows and 58 gCAP entries in S3, consistent with the methods; the 55-event caption is inconsistent. No supplied solution became an analysis input.

The user now expressly permits the external routine catalog for Figures 1A, 1C and S5, and for the paper's documented duration-magnitude regularization. Those roles are implemented separately from picking and association. Released trained PhaseNet weights, generic method software, original observations and required metadata are permitted. The Tan-fitted 1D model and station corrections, event results, picks and study-specific scripts remain excluded. No figure was digitized or used as numerical input.

## D2 Data access and actual coverage

| Quantity | Observed result |
|---|---|
| GNSS observations in plotted interval | 8743 |
| GNSS extension in 2025 (days) | 79 |
| GNSS overlap RMS after offset (m) | 0.000295 |
| All historical routine records | 26425 |
| Located historical routine records | 16046 |
| Located routine study events | 12503 |
| Difference from published study count | 420 |
| Processed waveform files | 318 |
| Processed union channel-hours | 2955.824 |

| Quantity | Observed result |
|---|---|
| Merged picks | 63036 |
| Provisional GaMMA events | 740 |
| Downloaded six-hour chunks | 144 |
| Pending six-hour requests | 74301 |

The study interval is 21 January 2022 through 20 March 2025, with the exclusive endpoint 21 March. The previous full-span HTTP 413 responses were request-size failures, not evidence of unavailable observations. The current collector queries actual FDSN availability and requests six-hour intervals with 30-second processing margins. Every successful miniSEED response is read, identified by network/station/channel, clipped to its core and hashed. Overlapping channel intervals are merged before computing channel-hours. The full queue contains 74,445 requests; unattempted and failed requests never count as coverage. The portable checkpoint and per-request receipts distinguish completed downloads from the pending multiyear workload. At this checkpoint the full period has not been retrieved or processed.

The collector resumes from `inputs/continuation/acquisition.sqlite`, uses six concurrent transfers and stops if shared free space falls below 20 GiB. Linear extrapolation from this download batch gives roughly 970 GB for the entire queue, against about 152 GiB free when checked. That estimate varies with compression and coverage and is not a measured full-archive size. Streaming processing with separate retention or additional storage is needed for the whole queue; pending work is labeled unfinished acquisition, not remote inaccessibility. Processing hours refer to successfully read input channels. Short fragments and missing components can prevent PhaseNet output, and these hours are not claimed as complete three-component detection exposure. The phase-classification warnings are retained in execution logs.

Fresh response-level StationXML was obtained for 22 IV stations and four 2I stations, including historical epochs. Public IT response queries returned HTTP 204. Eight additional IV identifiers missing from the XML—V0106, CDOV, CROS, CASE, V0102, V0105, V0104 and CMDP—were checked individually for full-period availability; all returned HTTP 204. These are service-specific no-data responses, not proof that local or author-held records do not exist. RAN's login page explicitly requires registration or credentials to download SAC/ASCII waveforms; those credentials are unavailable. This is an access limitation, not a prompt restriction. The previously accessed local/temporary archive supplies event-selected windows; those windows do not establish continuous exposure and are preserved as such. Exact actual source files are listed in the frozen input and processing manifests. Per-channel response-epoch intersections and every unobserved interval are saved separately; a usable response requires instrument sensitivity and response stages. This verifies metadata availability, not laboratory calibration accuracy. Channel/epoch coordinates differ slightly for some station identifiers; their spread is saved in `coordinate_variability.csv`. Association uses one static station coordinate per identifier, an additional approximation that matters for precise relocation.

The original RITE daily file and format note from Zenodo record 17132077 are readable. They contain MJD, Up displacement and uncertainty in meters, with 8,664 observations through 31 December 2024. No JPG was downloaded as an analysis input. A second INGV release, Zenodo 22259337, published 2 September 2026, contains seven-column observations and format notes through 2025. Its RITE file has 9,025 observations, ending 31 December 2025. We use its 79 daily values through 20 March 2025 to complete the study-period extension. Published MD5 checksums and local SHA-256 hashes were verified. Earlier searches of the Nevada service returned HTTP 404; the public RING RITE position file was readable but ended in 2016 despite its 2025 release timestamp. Neither failure is now treated as a remaining 2025-data blocker.

Routine HTML pages for every year 2005–2025 were retrieved on 12 October 2026. The selected historical interval contains 26,425 records, including unlocated events, and 16,046 located events. The study interval contains 19,274 routine records and 12,503 with latitude, longitude and depth. All five $M_d \geq 4$ events in the Figure 1 description are present. The 420-event excess over the paper's 12,083 located study events is unresolved: archive revisions and undocumented snapshot/selection differences may contribute, but their separate effects cannot be measured without the original historical snapshot. Counts were not tuned. A compressed source snapshot preserves the exact retrieved HTML because live pages can change.

## D3 Independent processing and explicit departures

### Displacement and routine-catalog context

The original Del Gaudio et al. table supplies 128 leveling entries, March 1905–July 2009. Display dates use the middle of the stated month; uncertain months remain flagged. The original March 1905 elevation is 3.464 m. Subtracting it yields a displacement range of −1.266 to 2.541 m; adding the same observed datum back supplies the ordinate used in the new figure. This datum comes from the numerical source table, not from published image pixels. Source notes identify entries estimated historically from photographs or earlier graphical reconstructions; those notes remain in the saved table. This project digitizes no graph.

We align daily GNSS observations to leveling using the median of 23 overlapping epoch offsets, each based on GNSS within 15 days of the leveling date. The offset is 1.6115 m relative to the 1905 displacement zero, with a 0.0025 m median absolute deviation. The 2025 release is aligned using all 8,664 overlapping daily values; its offset is 1.78951 m and overlap RMS after the shift is 0.000295 m. The plotted data contain 319 missing calendar days, which are not filled. Daily continuity is broken across gaps longer than seven days. The present 2025 endpoint is 3.0504 m above the 1905 datum. This assumes comparable displacement at the leveling benchmark and GNSS site; the paper's exact joining recipe is not specified.

Routine selections follow the written Figure 1 and S5 captions. Figure 1A uses located events from 1 January 2005 through 20 March 2025; stars mark $M_d \geq 4$. Figure 1C uses all routine events for monthly counts and strictly $M_d > 0.2$ for the cumulative count. The caption specifies this cutoff only for the cumulative curve. Retaining unlocated records in the monthly bins is therefore an explicit selection assumption; the paper does not specify their treatment, and no filter was chosen to match the published bars. The cumulative curve uses actual event times, avoiding a one-month bin-position shift. March 2025 includes only days 1–20. Figure S5 uses located events from 21 January 2022 onward. There is no additional tuned quality or geographic filter. Coordinates are a WGS84 azimuthal-equidistant projection centered at 14.14°E, 40.82°N; depth is below sea level. Natural Earth coastlines replace the unavailable original relief and inland zero-elevation contour. Marker-area scaling is an explicit display choice. The routine catalog never enters the inference or association code.

### Velocity inputs

The exact Calò–Tramelli (2018) association profile was searched in the readable publisher article and supplement, the article's data statement, INGV's Earth-prints repository, the permitted method-artifact tree, and publications/datasets citing the model. Earth-prints lists only the article PDF in its original-file bundle. Supplement section 2 describes Vp inherited from earlier studies and a depth-dependent Vp/Vs profile revised using Wadati diagrams; Figure S1.2 is graphical, without a numerical layer table. Its data statement directs requests to INGV. No graph was digitized and no Tan-fitted model was substituted.

A numerical alternative is verified in Tramelli et al. (2021), Scientific Reports 11, 7211, Table 1. Layer tops are 0, 0.5, 1, 3 and 10 km below sea level; Vp is respectively 1.7, 2.0, 3.0, 5.5 and 5.7 km/s. Vs is Vp/1.78. Layers are piecewise constant; the top value is extended above sea level for station elevations. Near-duplicate boundary nodes preserve discontinuities when preparing the 0.1 km eikonal grid. This is a published alternative, not a verified copy of the 2018 starting profile. All affected association/location results remain partial.

The permitted pinned method artifact `models/3d_vmod` is attributed in its README to Battaglia et al. (2008). It has a $23 \times 14 \times 27$ grid containing Vp followed by Vp/Vs, with Vs calculated by division. Original 2008 byte identity was not independently verified. Regional averages use explicitly assumed longitude/latitude boxes: caldera $[14.02, 14.20] \times [40.77, 40.86]$ and central $[14.11, 14.16] \times [40.81, 40.84]$. Above-sea-level air cells are excluded from the crustal starting average. The VELEST surface layer is extended upward; its next boundary is placed at +0.05 km to keep receivers inside the first layer as required by the generic manual. Tan's fitted 1D model and fitted station corrections are not read.

To resolve the exact-profile blocker, request the numerical layer-top depths, Vp and Vs or depth-dependent Vp/Vs underlying Calò–Tramelli supplementary Figure S1.2a/b, both layered and interpolated conventions, units/depth datum, and confirmation of the version used by Tan's GaMMA-1D. The data statement names INGV; Anna Tramelli at INGV–Osservatorio Vesuviano or corresponding author Marco Calò (`calo@igeofisica.unam.mx`) can identify the table. No one was contacted. The full search record is `docs/velocity_profile_search.json`.

## Picking, association and calibration

The original released PhaseNet weights are loaded with attribution through SeisBench. Each input is demeaned and high-pass filtered at 1 Hz with a causal four-pole Butterworth filter. Thirty-minute cores have 30-second margins. Both pick thresholds are 0.3; S picks at CASE, CFB1 and CFB3 are excluded. Arrivals within 0.25 s of a higher-probability pick on the same station/phase are merged. Gaps are split, not interpolated. The retraining demonstrations remain unperformed because the exact local annotations, splits and augmentations are unavailable.

GaMMA uses a Bayesian Gaussian mixture, 1D eikonal travel times, at least six picks, and no separate minimum P/S count. Unspecified settings are explicit assumptions: time-only likelihood, DBSCAN epsilon 10 s, minimum samples 3, oversampling factor 5, variance prior 1 $s^2$, horizontal bounds ±12 km, and depth 0–8 km. Independent clusters run on six CPU workers and saved identifiers are sorted by origin time. Association is partitioned by UTC day, so phases straddling midnight may be split; this is an additional boundary limitation. All input and coverage margins are clipped to the stated study interval. The first sequential association attempt was restarted with parallel clusters to reduce runtime; neither configuration uses a reference catalog to generate events.

The independent VELEST pilot completed 9 runs and retained 122 events after the final filters. The small calibration subset and absent independent phase review do not establish a reliable regional velocity model. The fitted layer $V_p/V_s$ ratios range from 1.214 to 2.076. The two regional starting choices yield RMS differences of 0.248 km/s in $V_p$ and 0.304 km/s in $V_s$. These sensitivities limit geological interpretation. Repeating calibration with 64-bit rather than 32-bit arithmetic changes some fitted layers by up to 0.48 km/s in $V_p$ and 0.60 km/s in $V_s$. This substantial arithmetic sensitivity prevents accepting the pilot fit as a robust regional calibration. The pilot geometry screen requires more than six picks and azimuthal gap below 180°, following generic VELEST guidance; it is not claimed to recreate the paper's manually selected 373 events. Nonpositive or invalid initial propagation times are recorded and removed before inversion. The geometry screen is recomputed; events with fewer than four remaining picks cannot proceed to the next residual-rejection solve. These are explicit numerical departures. Controls use seven iterations, hypocenter/origin damping 0.01, velocity damping 1, station

damping 0.1, and S/P observation weight 0.5. Two regional starting models and two calibration passes are attempted, followed by fixed-model catalog passes and the paper's 0.4 s/0.3 s residual rejection and final six-pick, 0.25 s RMS, 300° gap filters. If a pass fails or produces an unusable result, it is recorded rather than replaced by the authors' fitted solution. The controls, logs and candidate selections are saved; existence of code is not counted as successful inversion.

The expanded VELEST input exceeded the generic source's 658-event array capacity with 708 eligible events. Increasing the declared capacity to 1,024 and enabling array bounds checks removed the memory fault. The next attempt stopped at the Cholesky positive-definiteness check in 32-bit arithmetic. The current build uses 64-bit arithmetic, preserving the equations and damping. Calibration differences between precisions, the actual whole-catalog outcome and both failures are saved. These are implementation and numerical limitations, not missing stations; the station warning emitted during memory corruption was misleading.

The current magnitude pilot uses stations with at least one associated phase for each event; it has not yet measured every recorded but unassociated station. Magnitude measurements use picked P arrivals where available, otherwise independent predicted arrivals, a 5 s signal window, a 2 s preceding noise window and a 5 dB power-SNR threshold. Responses are removed to velocity with a stated prefilter and water level. Event separation must be less than one fifth of the common station distances, with at least two common stations. Channel log amplitude ratios are weighted by squared, arrival-aligned zero-lag correlation, an assumption where the precise correlation/weight specification is absent. Weighted least squares uses unit-weight routine-Md priors only for unique post-detection temporal matches within 2 s; multiple independent events claiming the same reference are rejected as ambiguous calibration anchors, without removing detections. Unconnected or unanchored components receive no magnitude. These priors are calibration inputs, not an independent validation set. Provisional initial locations limit any pilot magnitudes. The saved maximum-curvature/maximum-likelihood calculation cannot establish full-period completeness from this noncontinuous sample.

The CAAM/COLB hybrid scan uses the three written-method template times, 8 s templates, 0.2–6 Hz filtering, a two-station mean normalized correlation threshold of 0.25 and 8 s peak separation. Common returned intervals are intersected; duplicates and ten-second filter-edge regions are removed. It covers 59.757 hours and produces 702 unconfirmed threshold crossings. These are not confirmed hybrid clusters. The previous four-candidate review showed persistent fluctuations and did not establish isolated hybrid onsets. Cluster membership, three-year history, cluster-relative magnitudes and catalog-wide coda discrimination remain unfinished.

## D4 Supporting results and validation

The current run has 63036 merged picks and 740 provisional associations. Probability bounds, S-pick exclusions, the six-pick minimum, unique assignment and valid pick indices are checked. The previous homogeneous-medium association fixture recovered one synthetic event with 16 arrivals; it tests coordinate/association consistency, not detection accuracy. The retained C++ equation checks are rerun: All 27 checks passed. The maximum absolute FI error across the analytic mixtures was 4e-15; the maximum exact-recovery magnitude error was 1.94e-16. These errors compare against analytic fixtures, not observed earthquakes.

The present S–P check includes 1043 paired observations, of which 36 are nonpositive. There are 40 associated CPOZ P arrivals. These defects and the alternative velocity profile prevent treating raw associations as validated locations. Internal residuals from the same model do not establish absolute accuracy. The location-check tables retain failed observations. The previous generic/retrained comparison on an unlabeled 1600 s CMIS window produced 17 P/4 S and 11 P/55 S picks respectively; the

120 s CSOB window produced zero and one pick. Those counts remain supporting evidence only, not training or labeled-accuracy reproduction, and are not inserted in S4. An analytic vertical-ray check of the published alternative gives maximum P/S errors of 0.040642/0.072342 s on the 0.1 km eikonal grid; halving the spacing reduces them to 0.020321/0.036171 s. This confirms first-order interface discretization behavior, not the physical accuracy of the alternative profile. The numerical checks and refinement results are saved in `data/continuation/profile_ray_checks.csv`.

The current amplitude/SNR stage measured 12473 channel amplitudes, with 2043 passing SNR and 377 accepted difference equations. It returned 22 provisional magnitudes. These are actual executed outputs, not a promised future count. Candidate measurement errors and quality failures are retained in the JSON/CSV records. The old 140-measurement, 23-SNR, zero-edge result is superseded for the current sample but preserved in Git and the earlier data directory.

### Comparison after independent processing

The conditional magnitude-frequency fit gives $M_c = 0.3$ and $b = 1.24 \pm 0.61$ from only 8 selected magnitudes at or above the pilot completeness level. The quoted Shi–Bolt standard error excludes coverage, selection and magnitude-calibration biases. The previous 20-event reporting cutoff was an unnecessary implementation assumption and is removed. This calculation does not establish population completeness or a regional b-value.

A strictly downstream comparison against supplied Data S1 (54,319 events) finds 64 unique temporal matches within two seconds among 122 provisional events. For these matches, the median absolute origin-time difference is 0.235 s and median horizontal difference is 0.819 km (maximum 7.894 km). These differences describe a limited VELEST pilot without TomoDD or manual phase review. Temporal proximity alone does not prove event identity, and unmatched events are not declared false detections. The comparison changes no independent data or setting.

## D5 Limitations and unfinished analyses

No publication panel is fully independently reproduced. Partial counterparts: 1A, 1B, 1C, S1, S2, S5/map, S5/northing-depth, S5/easting-depth, S6/map, S6/northing-depth, S6/easting-depth, S8A, S8B, S15C. All remaining panels are not reproduced. Routine-context permissions and the 2025 displacement gap are resolved. The full continuous download/processing workload remains unfinished. Exact Calò–Tramelli layer values are an external-input limitation; the released fixed 3D artifact is available, with a provenance caveat. RAN credentials are a specific access limitation.

The regional calibrated and TomoDD-relocated catalog, four HypoDD SVD subsets, reliable multi-year magnitudes/completeness, manually reviewed polarities and S/P ratios, SKHASH/gCAP solutions, FK Green's functions, exact event-specific gCAP filters/windows/weights, relocated CPOZ sector gathers, manual deep-event review and three-year hybrid clustering are not complete. Generic TomoDD source has now been retrieved, classified and compiled; compilation alone is not relocation. Training labels/splits and some inversion settings are incompletely specified. Supplied author-derived results could fill plots only by violating the independence restriction, so they are not substituted. That restriction is distinct from unfinished implementation or genuinely missing source inputs.

## D6 Figure-by-figure record

The machine-readable `docs/panel_record.json` records every panel's written specification, parameters, exact input paths, executed data/plot commands, saved products and limitation. The raw-waveform

manifest is `docs/continuation_processed_inputs.json`; its intervals are a frozen checkpoint, not the full queue. Context panels use `continue_context.py` and `plot_continuation.py`. S1 uses `process_continuous.py` then `plot_continuation.py`. S2 and S15C preserve their independently generated source recipes. All unmatched comparison panels remain explicit placeholders.

| Panel/view | Quantity | Status, specification and limitation |
|---|---|---|
| 1A | INGV event map and large-event markers | partially reproduced. Main p.71, Fig.1 caption. 2005-01-01 to 2025-03-20; origin 14.14 E, 40.82 N; sea-level depth; Md>=4 markers. Current catalog snapshot differs from the historical snapshot: 12,503 located study events versus 12,083. No tuning to published counts. Natural Earth coastline substitutes for the unavailable original relief/inland-zero contour. |
| 1B | Vertical displacement at Pozzuoli port | partially reproduced. Main p.71, Fig.1 caption. 128 leveling observations, 1905–2009; daily GNSS 2000-05-29 through 2025-03-20. Median offset from 23 overlapping leveling epochs, +/-15 days. Extension beyond original 2023 endpoint is labeled.. Benchmark equivalence and datum-joining convention are assumed because the exact original stitching recipe is not specified. Calendar gaps remain unfilled; no missing recent-data blocker through 2025-03-20. |
| 1C | Temporal seismicity and cumulative counts | partially reproduced. Main p.71, Fig.1 caption. 2005-2025; cumulative Md>0.2; study interval 2022-01-21 to 2025-03-20. Monthly counts and exact-time cumulative Md>0.2 reconstructed from the current routine snapshot; historical count differences unresolved. March 2025 is a partial month. |
| 2A | Catalog seismicity rates by region | not reproduced. Main p.72, Fig.2 caption. Central vs ring/other regions; 54319 vs 12083 catalog events; time bin unspecified. 740 provisional GaMMA events are not a complete calibrated, TomoDD-relocated catalog. Multiyear acquisition and downstream processing remain incomplete. |
| 2B | Depth versus time, symbol size by magnitude | not reproduced. Main p.72, Fig.2 caption. Sea-level depth; near-surface 0-0.5 km; central 0.5-2.1; ring sectors to 3.7 km. 740 provisional GaMMA events are not a complete calibrated, TomoDD-relocated catalog. Multiyear acquisition and downstream processing remain incomplete. |
| 3/map | Epicentral spatial pattern | not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. 740 provisional GaMMA events are not a complete calibrated, TomoDD-relocated catalog. Multiyear acquisition and downstream processing remain incomplete. |
| 3/northing-depth | Depth versus northing | not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. 740 provisional GaMMA events are not a complete calibrated, TomoDD-relocated catalog. Multiyear acquisition and downstream processing remain incomplete. |
| 3/ easting-depth | Depth versus easting | not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. 740 provisional GaMMA events are not a complete calibrated, TomoDD-relocated catalog. Multiyear acquisition and downstream processing remain incomplete. |
| 4A | Ring-sector focal mechanisms and seismicity | not reproduced. Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. Independent calibrated locations, reviewed polarities and gCAP configurations are missing; focal inversions not run. Data S3 actually contains 74 rows/58 gCAP entries, consistent with methods; caption says 55. |

| Panel/view | Quantity | Status, specification and limitation |
|---|---|---|
| 4B | Central/onshore focal mechanisms and seismicity | not reproduced. Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. Independent calibrated locations, reviewed polarities and gCAP configurations are missing; focal inversions not run. Data S3 actually contains 74 rows/58 gCAP entries, consistent with methods; caption says 55. |
| S1 | Station data-availability timeline | partially reproduced. Supplement S12 caption; S2 methods. 48 stations; continuous-data availability. 318 files processed, 2955.824 union channel-hours. Full-period queue still pending; actual gaps and unavailable RAN credentials remain. Requests/availability are not coverage. |
| S2 | Station map | partially reproduced. Supplement S13 caption; S2 methods. 23 broadband, 9 short-period, 16 accelerometers. 51-station permitted metadata inventory differs from the paper's 48-station description; exact epoch/membership and basemap not reproduced. |
| S3A | Noise example: waveform, phase labels, spectrogram | not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only. |
| S3B | Single-event example: waveform, phase labels, spectrogram | not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only. |
| S3C | Two overlapping events: waveform, phase labels, spectrogram | not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only. |
| S3D | Five overlapping copies of one event: waveform, phase labels, spectrogram | not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only. |
| S4A | Swarms at CSOB: waveform, spectrogram, original and retrained picker scores | not reproduced. Supplement S15 caption; S2-S3. 120 s; 2 h after May 2024 Md4.4 event; phase threshold 0.3. Original and retrained models both executed on independently selected CSOB/CMIS calendar windows. Exact published windows and independent reference annotations remain unidentified; these diagnostics are not the published benchmark. |
| S4B | CMIS noise: waveform, spectrogram, original and retrained picker scores | not reproduced. Supplement S15 caption; S2-S3. 1600 s; coastal noise; phase threshold 0.3. Original and retrained models both executed on independently selected CSOB/CMIS calendar windows. Exact published windows and independent reference annotations remain unidentified; these diagnostics are not the published benchmark. |
| S5/map | Routine-catalog spatial views; map | partially reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. Current catalog snapshot differs from the historical snapshot: 12,503 located study events versus 12,083. No tuning to published counts. Natural Earth coastline substitutes for the unavailable original relief/inland-zero contour. |

| Panel/view | Quantity | Status, specification and limitation |
|---|---|---|
| S5/northing-depth | Routine-catalog spatial views; northing-depth | partially reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. Current catalog snapshot differs from the historical snapshot: 12,503 located study events versus 12,083. No tuning to published counts. Natural Earth coastline substitutes for the unavailable original relief/inland-zero contour. |
| S5/ easting-depth | Routine-catalog spatial views; easting-depth | partially reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. Current catalog snapshot differs from the historical snapshot: 12,503 located study events versus 12,083. No tuning to published counts. Natural Earth coastline substitutes for the unavailable original relief/inland-zero contour. |
| S6/map | Initial-location spatial views; map | partially reproduced. Supplement S17 caption. Pilot calibration uses 98 independent candidates, not the published 373. Two regional starts and two calibration rounds each; five fixed-model catalog passes; rejection at 0.4 then 0.3 s; final >=6 picks, RMS <=0.25 s, gap <=300 degrees. Positive-time filtering and geometric selection are explicit assumptions; dampings and boxes are in the implementation record.. Independent VELEST pilot: calibration and whole-catalog passes executed, but limited/noncontinuous sample and unreviewed input phases prevent an exact result. |
| S6/northing-depth | Initial-location spatial views; northing-depth | partially reproduced. Supplement S17 caption. Pilot calibration uses 98 independent candidates, not the published 373. Two regional starts and two calibration rounds each; five fixed-model catalog passes; rejection at 0.4 then 0.3 s; final >=6 picks, RMS <=0.25 s, gap <=300 degrees. Positive-time filtering and geometric selection are explicit assumptions; dampings and boxes are in the implementation record.. Independent VELEST pilot: calibration and whole-catalog passes executed, but limited/noncontinuous sample and unreviewed input phases prevent an exact result. |
| S6/ easting-depth | Initial-location spatial views; easting-depth | partially reproduced. Supplement S17 caption. Pilot calibration uses 98 independent candidates, not the published 373. Two regional starts and two calibration rounds each; five fixed-model catalog passes; rejection at 0.4 then 0.3 s; final >=6 picks, RMS <=0.25 s, gap <=300 degrees. Positive-time filtering and geometric selection are explicit assumptions; dampings and boxes are in the implementation record.. Independent VELEST pilot: calibration and whole-catalog passes executed, but limited/noncontinuous sample and unreviewed input phases prevent an exact result. |
| S7A/map | Cluster locations and 2-sigma errors: map and cross-sectional views; map | not reproduced. Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished. |
| S7A/ easting-depth | Cluster locations and 2-sigma errors: map and cross-sectional views; easting-depth | not reproduced. Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished. |
| S7B | EX uncertainty versus differential-pick count | not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished. |

| Panel/view | Quantity | Status, specification and limitation |
|---|---|---|
| S7D | EY uncertainty versus differential-pick count | not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished. |
| S7F | EZ uncertainty versus differential-pick count | not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished. |
| S7C | EX uncertainty density by cluster | not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished. |
| S7E | EY uncertainty density by cluster | not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished. |
| S7G | EZ uncertainty density by cluster | not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished. |
| S8A | Estimated local versus routine duration magnitude | partially reproduced. Supplement S19 caption; S5-S6 methods. 5 s signal, 2 s preceding noise; SNR >=5 dB; stations with associated phases only; >=2 shared stations; separation < one fifth of station distance; squared zero-lag correlation weights; Md regularization lambda=1 for unique one-to-one matches within 2 s. Initial locations. B uses 0.1 magnitude bins, maximum-curvature Mc, half-bin-corrected likelihood b and formal Shi-Bolt sampling error; no arbitrary minimum sample cutoff.. 12473 channel measurements, 2043 passing SNR, 377 relative-amplitude equations, 22 provisional magnitudes. Final relocated locations/full exposure remain missing; routine Md is an allowed regularizer, not independent validation. |
| S8B | Magnitude-frequency distribution and fitted b value | partially reproduced. Supplement S19 caption; S6 methods. 5 s signal, 2 s preceding noise; SNR >=5 dB; stations with associated phases only; >=2 shared stations; separation < one fifth of station distance; squared zero-lag correlation weights; Md regularization lambda=1 for unique one-to-one matches within 2 s. Initial locations. B uses 0.1 magnitude bins, maximum-curvature Mc, half-bin-corrected likelihood b and formal Shi-Bolt sampling error; no arbitrary minimum sample cutoff.. 12473 channel measurements, 2043 passing SNR, 377 relative-amplitude equations, 22 provisional magnitudes. Final relocated locations/full exposure remain missing; routine Md is an allowed regularizer, not independent validation. |
| S9/map | Filtered-catalog spatial views and feature labels A-D; map | not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S9/northing-depth | Filtered-catalog spatial views and feature labels A-D; northing-depth | not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |

| Panel/view | Quantity | Status, specification and limitation |
|---|---|---|
| S9/ easting-depth | Filtered-catalog spatial views and feature labels A-D; easting-depth | not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S10/2022 | Annual seismicity map | not reproduced. Supplement S21 caption. Year 2022. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S10/2023 | Annual seismicity map | not reproduced. Supplement S21 caption. Year 2023. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S10/2024 | Annual seismicity map | not reproduced. Supplement S21 caption. Year 2024. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S10/2025 | Annual seismicity map | not reproduced. Supplement S21 caption. Year 2025 through March20. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S11/swarm-1 | Map of one of four major swarms | not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S11/swarm-2 | Map of one of four major swarms | not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S11/swarm-3 | Map of one of four major swarms | not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S11/swarm-4 | Map of one of four major swarms | not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S12A | Ring-sector division | not reproduced. Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. 40 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy. |

| Panel/view | Quantity | Status, specification and limitation |
|---|---|---|
| S12B | Ring-sector division in complementary spatial view | not reproduced. Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. 40 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy. |
| S12C | Common-receiver gather: east / HHZ | not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 40 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy. |
| S12D | Common-receiver gather: southwest / HHZ | not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 40 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy. |
| S12E | Common-receiver gather: north / HHZ | not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 40 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy. |
| S12F | Common-receiver gather: east / HHN | not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 40 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy. |
| S12G | Common-receiver gather: southwest / HHN | not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 40 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy. |
| S12H | Common-receiver gather: north / HHN | not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 40 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy. |
| S13A | Full caldera seismicity on satellite reference | not reproduced. Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |
| S13B | Onshore seismicity on satellite reference | not reproduced. Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted. |

| Panel/view | Quantity | Status, specification and limitation |
|---|---|---|
| S14/1 | CAAM waveforms and event annotations | not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. 59.757 common CAAM/COLB scan hours; 702 unconfirmed threshold crossings. Three-year scan, reviewed clusters, relative magnitudes and timing histories remain unfinished. |
| S14/2 | CAAM waveforms and event annotations | not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. 59.757 common CAAM/COLB scan hours; 702 unconfirmed threshold crossings. Three-year scan, reviewed clusters, relative magnitudes and timing histories remain unfinished. |
| S14/3 | CAAM waveforms and event annotations | not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. 59.757 common CAAM/COLB scan hours; 702 unconfirmed threshold crossings. Three-year scan, reviewed clusters, relative magnitudes and timing histories remain unfinished. |
| S14/ unclustered | CAAM waveforms and event annotations | not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. 59.757 common CAAM/COLB scan hours; 702 unconfirmed threshold crossings. Three-year scan, reviewed clusters, relative magnitudes and timing histories remain unfinished. |
| S15A | Hybrid family 1: waveforms and spectrograms | not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Post-generation visual comparison shows a different published example from the written-method template used here. The generated family-template figure is related analysis, not reproduction of this panel. |
| S15B | Hybrid family 2: waveforms and spectrograms | not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Post-generation visual comparison shows a different published example from the written-method template used here. The generated family-template figure is related analysis, not reproduction of this panel. |
| S15C | Hybrid family 3: waveforms and spectrograms | partially reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Written template times used; exact published example alignment, display filtering and spectral settings incompletely specified. Sensitivity-corrected velocities and stated independent display settings used. |
| S15D | Shallow volcano-tectonic example: waveforms and spectrograms | not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Exact shallow VT example not identified by written method alone; no author catalog or plot values used to select it. |

# D7 Saved numerical tables and equation consistency checks

## Original-waveform template frequency indices

These retained numerical measurements describe written-method templates, not confirmed hybrid families. E1 and E2 are not substituted for the different published S15A/B examples.

| Template | FI | Samples |
|---|---|---|
| E1 | -0.7862 | 800 |

| Template | FI | Samples |
|---|---|---|
| E2 | -0.4286 | 800 |
| E3 | -1.0819 | 800 |

### Unvalidated pilot velocity fit

These values come from the present independent pilot and are supporting evidence only, not an accepted reproduction of Table S1. Positive velocities, increasing layer interfaces and $V_p > V_s$ are checked; these tests do not establish resolution or accuracy.

| Layer top (km b.s.l.) | $V_p$ (km/s) | $V_s$ (km/s) | $V_p/V_s$ |
|---|---|---|---|
| -0.500 | 1.730 | 1.190 | 1.454 |
| 0.050 | 1.890 | 1.180 | 1.602 |
| 0.100 | 1.890 | 1.160 | 1.629 |
| 0.250 | 1.940 | 0.970 | 2.000 |
| 0.500 | 1.730 | 1.230 | 1.407 |
| 0.750 | 2.080 | 1.040 | 2.000 |
| 1.000 | 2.470 | 1.190 | 2.076 |
| 1.250 | 2.490 | 1.440 | 1.729 |
| 1.500 | 2.820 | 1.420 | 1.986 |
| 1.750 | 3.120 | 1.640 | 1.902 |
| 2.000 | 2.900 | 2.030 | 1.429 |
| 2.500 | 3.980 | 2.780 | 1.432 |
| 3.000 | 4.290 | 3.440 | 1.247 |
| 3.500 | 5.450 | 3.170 | 1.719 |
| 4.000 | 4.930 | 4.060 | 1.214 |
| 4.500 | 5.300 | 3.450 | 1.536 |
| 5.000 | 5.550 | 3.540 | 1.568 |
| 5.500 | 5.810 | 3.470 | 1.674 |
| 6.000 | 5.960 | 3.540 | 1.684 |
| 6.500 | 6.100 | 3.590 | 1.699 |
| 7.000 | 6.230 | 3.590 | 1.735 |

### Analytic fixtures

| Check | Absolute error | Tolerance |
|---|---|---|
| equal_amplitudes | 0 | 1e-14 |
| decade_amplitude | 0 | 1e-14 |
| ratio_reversal | 0 | 1e-14 |
| common_station_gain | 5.55e-17 | 1e-14 |
| zero_amplitude_rejected | 0 | 0 |
| negative_power_rejected | 0 | 0 |
| fi_mixture_0 | 4e-15 | 1e-10 |
| fi_mixture_1 | 8.88e-16 | 1e-10 |
| fi_mixture_2 | 1.78e-15 | 1e-10 |

| Check | Absolute error | Tolerance |
|---|---|---|
| fi_mixture_3 | 1.55e-15 | 1e-10 |
| fi_mixture_4 | 1.78e-15 | 1e-10 |
| fi_mixture_5 | 2.11e-15 | 1e-10 |
| fi_mixture_6 | 2.22e-16 | 1e-10 |
| fi_mixture_7 | 2.11e-15 | 1e-10 |
| fi_mixture_8 | 1.33e-15 | 1e-10 |
| fi_mixture_9 | 8.88e-16 | 1e-10 |
| fi_mixture_10 | 1.33e-15 | 1e-10 |
| fi_mixture_11 | 4.44e-16 | 1e-10 |
| fi_mixture_12 | 1.78e-15 | 1e-10 |
| parseval | 3.33e-15 | 1e-10 |
| fi_circular_shift | 7.22e-16 | 1e-10 |
| fi_gain | 8.88e-16 | 1e-10 |
| snr_5s_2s_windows | 8.88e-16 | 1e-10 |
| magnitude_exact_recovery | 1.94e-16 | 1e-12 |
| magnitude_pair_order | 0 | 1e-12 |
| magnitude_gauge_shift | 4.44e-16 | 1e-12 |
| unanchored_problem_rejected | 0 | 0 |

## D8 Rebuild and replacements

The clean continuation procedure is documented in `docs/CONTINUATION.md`. Restore the frozen inputs with `scripts/restore_inputs.py--continuation`; run `scripts/rebuild_continuation.py--fresh` using the repository's Python environment. The rebuild regenerates observations, picking/association, the downstream attempts, context panels and both PDFs. It is a rebuild of the frozen observed subset, not proof of complete multiyear reproduction. Its actual outcome and executed commands are saved in `data/continuation/rebuild_status.json` and `data/continuation/rebuild_log.json`. The first full attempt restarted after fresh picking to include explicit study-boundary clipping. A subsequent amplitude attempt failed with a terminated worker process; the retry uses spawned processes and hash-verified resumption of the completed stages. These are recorded implementation failures, not unavailable input data. The document build verifies main-text sections, Figures 1–4, supplement S1–S15, and the nineteen comparison pages in order, and confirms that published images occur only in the comparison PDF.

This continuation replaces the prior report/comparison PDFs and generated figure fragments, panel records and obsolete context figure usage. The old displacement display ending in 2019 is replaced by observed daily coverage through March 2025. The old context plots are superseded by the corresponding Figure 1 and S5 outputs; their scripts/data remain traceable, while their obsolete PDFs/PNGs are removed after the current counterparts exist. New seismic outputs are in `data/continuation`, preserving the old numerical baseline in `data/extended`. Exact changes are documented in `docs/REPLACEMENTS.md`. Project-specific caches remain under `tmp`; pip's normal cache is `/home/mehffin/.cache/pip`. Large waveforms and model/software artifacts remain ignored under `inputs`, with retrieval URLs, file sizes and hashes.

## D9 Sources

The primary specification is Tan et al. (2025), Science 390, 70–75, local `tanmodel.pdf` and `tanmodelsupp.pdf`. Observation/metadata services: https://webservices.ingv.it/fdsnws/; routine catalog: https://terremoti.ov.ingv.it/gossip/flegrei/. GNSS: De Martino and colleagues, https://zenodo.org/records/17132077 and https://zenodo.org/records/22259337, with the requested citation to De Martino et al. (2021), doi:10.3390/rs13142725. Historical leveling: Del Gaudio et al. (2010), doi:10.1016/j.jvolgeores.2010.05.014. Published alternative profile: Tramelli et al. (2021), doi:10.1038/s41598-021-86506-6, Table 1. Requested exact profile: Calò and Tramelli (2018), doi:10.1038/s41598-018-34456-x. Fixed external grid attribution: Battaglia et al. (2008), doi:10.1111/j.1365-2478.2007.00687.x.

The permitted artifact revision is

`f84c9c9dc6bb1501a95a833f6634199a63d4f3b3`.

GaMMA revision is

`80394dd4a4c29450da599f507916ecf1db6203dd`.

Generic VELEST, HypoDD and TomoDD source receipts remain in the repository. No Tan-specific analysis or plotting script was used.